Elucidating the molecular mechanisms for the interactions between N. meningitidis and human cells

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INTRODUCTION:

Neisseria meningitidis, in its virulent variant, can pass through human endothelial cells by a species-specific mechanism to reach the brain membranes, causing meningitis. The molecular mechanism, by which the pathogen enters the human host, is caused by an initial interaction between the meningococcal type IV pilus (Tfp) and the CD147 membrane receptor of human cells [1].

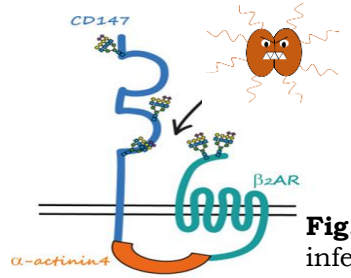

Fig.1 Meningococcal infection process [2]

COMPUTATIONAL STRATEGY:

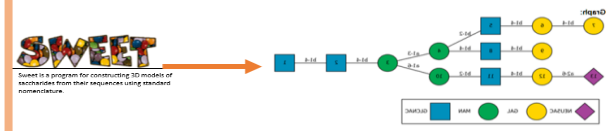

Fig.2 Based on the experimental data of Le Guennec and his colleagues [3], glycan was built with a computational metod.

GlyProt

In silico glycosylation of protein

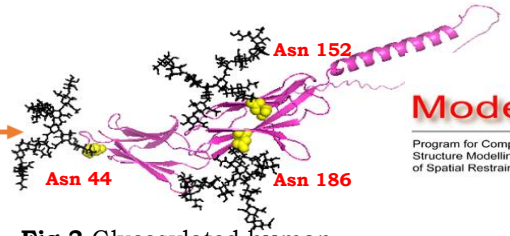

Fig.3 Glycosylated human CD147 receptor model.

Modeller

Program for Comparative Protein Structure Modelling by Satisfaction of Spatial Restraints

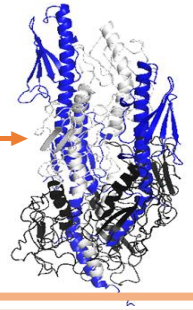

Fig. 4 Model of the surface portion of Tpf, containing whole pilinE (in white), pilinE heads (in black) and pilinV (in blue).

PATCHDOCK

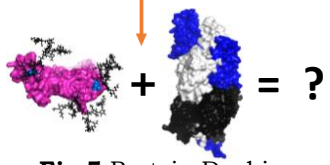

Fig.5 Protein Docking

RESULTS:

ACE= -361,82 Kcal/mol

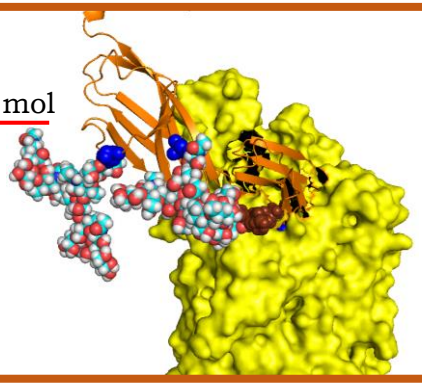

Fig. 6 PatchDock interaction model with Tfp (in yellow), human CD147 (in orange), Neu5Ac of glycan on Asn 186 (in brown).

ACE= -375,97 Kcal/mol

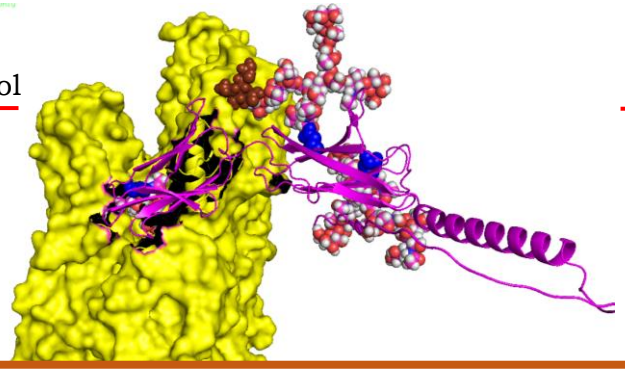

Fig. 7 PatchDock interaction model with Tfp (in yellow), human CD147 (in magenta), Neu5Ac of glycan on Asn 186 (in brown).

ACE= -442,12 Kcal/mol

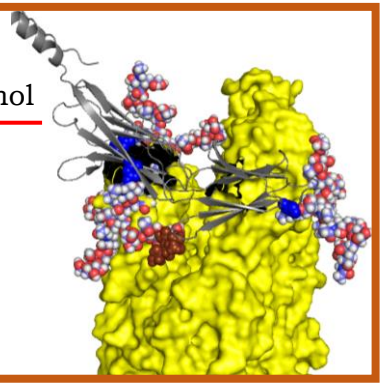

Fig. 8 PatchDock interaction model with Tfp (in yellow), human CD147 (in grey), Neu5Ac of glycan on Asn 186 (in brown).

FUTURE DEVELOPMENT: Further analyses are underway to better understand the nature of interactions.

REFERENCES: [1] Bernard SC et al., Nat Med. 2014 Jul;20(7):725-31.; [2] Marullo S. et al., 2022 May 2;13:883568. doi: 10.3389/fendo.2022.883568.; [3] Le Guennec L. et al., Proc Natl Acad Sci U S A. 2020 Feb 4;117(5):2606-2612.